

My grades for midterm



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1) (12 marks) You are to draw an Entity-Relationship diagram that shows the following information relating to a publisher of non-fiction books.

For books, the ISBN (which is a unique number), title, topic and intended publication date are to be recorded.

For authors, the (unique) SIN, first name, last name, date of birth and nationality are to be recorded.

Books are (obviously, I hope) written by authors. Books may have more than one author but must have at least one and authors often write more than one book. The DB is to include information about prospective authors (who have not yet written any books). The date that each author started work on a book is to be recorded.

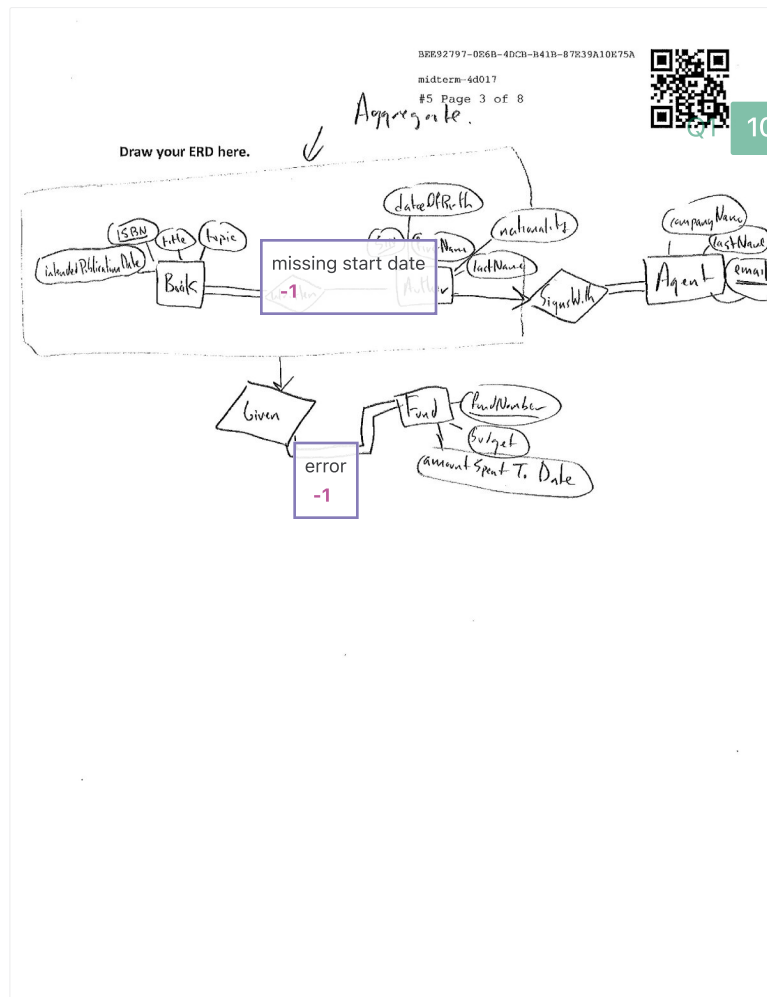
One of the major purposes of the DB is to keep track of funds that are granted to authors writing books. Each fund has a unique number, a budget and the amount spent to date. The publisher provides a separate fund for each author when they work on a book. An author can have a maximum of one fund for any one book (and may not be given a fund) but can have other funds for different books. There are no funds that are not associated with an author and book.

For example, assume that C. Darwin, P. King and R. FitzRoy co-authored the book *Voyage of the Beagle* (*Voyage*). There may be as many as three funds associated with *Voyage*, one for each author. If the authors write other books, they may also have separate funds associated with those books. So, C. Darwin might have a second fund related to the book *On the Origin of Species*.

Finally, the DB is to keep track of authors agents. The company name, last name and email addresses of agents are to be recorded. Every agent has an email address which is unique.

An author can have at most one agent (and may not have one). One agent can represent many authors and the DB will not record information about agents who are not associated with any authors.

Draw the ERD for the situation described above (on the next page of the exam paper).





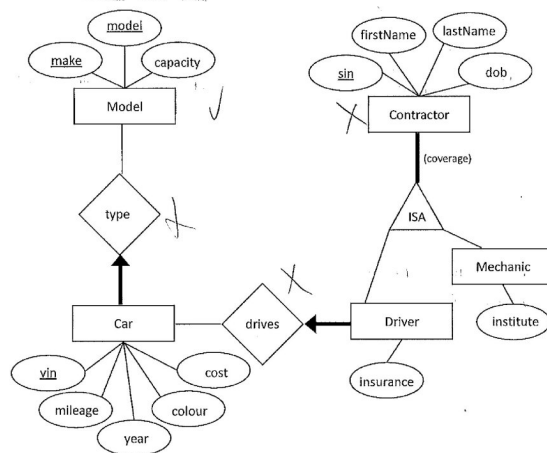
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2) (12 marks) Write SQL create table statements for the *type* relationship set and the *driver* entity set in the ERD shown below. You should:

- Represent participation and key constraints where possible.
- Assume that a separate table for *drives* is **not** to be created, even though *type* is to be a separate table.
- A separate table for *Contractor* is also **not** to be created.
- Attribute domains are:
 - make, model, colour, firstName, lastName, institute – char length 20
 - sin – char, length 11
 - vin, insurance – char, length 9
 - year, dob – date
 - capacity, mileage, cost – real



Reminder – you are to write create table statements for just the two specified sets (*type* and *driver*) and **not** the entire ERD!

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2

Answer Question 2 here

```
CREATE TABLE Model (
  make CHAR(20),
  model CHAR(20),
  capacity REAL,
  PRIMARY KEY (make, model)
)
```

Type?

missing table

-6

```
CREATE TABLE Car (
  make CHAR(20), vin CHAR(9),
  model CHAR(20), mileage REAL,
  year DATE,
  color CHAR(20),
  cost REAL,
  CONSTRAINT fk_model FOREIGN KEY (make, model) REFERENCES Model
  PRIMARY KEY (vin)
)
```

many-to-one relationship

```
CREATE TABLE Driver (
  insurance CHAR(9),
  sin CHAR(11),
  CONSTRAINT fk_contractor FOREIGN KEY (sin) REFERENCES Contractor
  ON DELETE CASCADE,
  PRIMARY KEY (sin)
)
```

missing attribute(s)

-1 missing attribute(s)

-1

not null

NOT NULLS

wrong attribute

-1

```
CREATE TABLE Mechanic (
  institute CHAR(20),
  sin CHAR(11),
  CONSTRAINT fk_contractor FOREIGN KEY (sin) REFERENCES Contractor
  ON DELETE CASCADE,
  PRIMARY KEY (sin)
)
```



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3) (12 marks) This schema represents a DB to store information about music. Primary keys are underlined, and foreign keys are shown in superscript.

- Band = (bandName, members, genre, country)
- Musician = (sin, firstName, lastName, instrument)
- Plays = (bandName^{band}, sin^{musician}, share)
- Song = (isrc, bandName^{band}, title, year, duration, album)

Write **relational algebra** queries to return the information set out below.

a) isrc and title of songs recorded by Deep Purple before 1980

$\pi_{isrc, title} (\sigma_{bandName = 'Deep Purple' \wedge year < 1980} (Songs))$

full marks
3

Q3a 3

b) First and last names of musicians who play guitar (an instrument) in a rock band (a genre)

$\pi_{firstName, lastName} (\sigma_{instrument = 'guitar' \wedge genre = 'rock'} (Musician \bowtie Plays \bowtie Band))$

full marks
3

Q3b 3

c) First and last names and band names of musicians who have played in the same bands as any musician named Jeff Beck.

$\pi_{firstName, lastName, bandName} (\sigma_{firstName = 'Jeff' \wedge lastName = 'Beck' \wedge \exists ba: ba \neq 'Jeff' \wedge ba \neq 'Beck' \wedge ba \bowtie Plays \bowtie Band} (Musician \bowtie Plays))$

full marks
3

Q3c 3

d) Song titles of songs recorded by The Beatles (a band name) that are not also the titles of songs recorded by any other UK (a band country) band.

$\pi_{titles} (Songs - \sigma_{bandName = 'The Beatles'} (Songs))$

no marks
0

Q3d 0

Use cartesian product.

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4) (9 marks) Write SQL queries to return the information set out below.

- Band = {bandName, members, genre, country}
- Musician = {sin, firstName, lastName, instrument}
- Plays = {bandName^{band}, sin^{musician}, share}
- Song = {isrc, bandName^{band}, title, year, duration, album}

Song title
↓

a) Isrc, song names and band names of songs that contain the word love or Love.

SELECT S.isrc, S.title, S.bandName
FROM Song S
WHERE S.title = 'Love'

full marks
3

Q4a 3

b) Song titles and years of songs recorded by any band that had Ian Gillan (a musician) in it.

SELECT S.title
FROM Musician M, Plays P, Song S
WHERE M.firstName = 'Ian' AND M.lastName = 'Gillan'
AND M.bandName = P.bandName AND P.bandName = S.bandName

minor error
2

Q4b 2

c) Musician SINS, last names and instruments of musicians who have played in both metal and country bands (metal and country genres).

SELECT M.sin, M.lastName, M.instrument
FROM Musician M, Plays P, Band B
WHERE M.sin = P.sin AND P.bandName = B.bandName AND B.genre = 'metal'
INTERSECT
SELECT M.sin, M.lastName, M.instrument
FROM Musician M, Plays P, Band B

full marks
3

Q4c 3

WHERE M.sin = P.sin AND P.bandName = B.bandName AND B.genre = 'country'



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